Toward Fine-Grained Remote Sensing Image–Text Retrieval via Hierarchical Multigranularity Alignment

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***Abstract*—Remote sensing text–image retrieval (RSTIR) has attracted increasing attention from researchers because of advancements in multimodal foundation models and the growing volume of remote sensing (RS) data. The mainstream approaches involve adapting pretrained models to the RS domain through transfer learning. However, existing methods often overlook fine-grained correspondences between images and textual descriptions, making it challenging to effectively align key regions in images with corresponding fine-grained textual semantics. We identify two primary limitations of mainstream methods: the inherent limitations of model capabilities and the scarcity of fine-grained datasets. To address these challenges, we propose an innovative fine-grained text–image retrieval (FTIR) method, along with a high-quality fine-grained dataset named RSITMD-FG. First, we design a region-query association module that enhances the model's ability to perceive fine-grained semantics by constraining the alignment between key regions in images and keywords in textual queries during training. Second, we propose a multigranularity alignment strategy to mitigate the difficulty and noise sensitivity of fine-grained alignment. Our proposed alignment strategy transitions from coarse-grained context semantics to fine-grained word-level semantics. Finally, we construct the RSITMD-FG dataset that emphasizes important region features within images and includes more fine-grained textual queries. To the best of our knowledge, RSITMD-FG is currently the most fine-grained publicly available dataset for RSTIR. We validate the effectiveness of our method on different benchmarks. Without complex multimodal interactions or additional training data augmentation, our concise method achieves competitive performance, outperforming even some domain-specific models. Our code and dataset will be released at https://github.com/ZhangWeihang99/FTIR**

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I. Introduction

RECENTLY, the rapid growth of remote sensing (RS) data has significantly expanded the scope of Earth observation [1], [2], [3]. Along with continuous advances in data processing techniques, the growth of RS data has supported a wide range of applications, including ecological monitoring [4], [5], land-use planning [6], [7], and disaster forecasting [8], [9]. Improving the efficiency of these applications relies heavily on the support of RS image retrieval technologies [10]. RS text–image retrieval (RSTIR) has become a fundamental task in the field of RS vision-language research and has attracted increasing attention from the research community [11], [12], [13]. RSTIR methods integrate both visual and linguistic understanding, which can retrieve images that best match a textual query or retrieve descriptions that most accurately represent a visual query.

The mainstream advanced approaches can be broadly classified into two categories: directly transferring knowledge from pretrained models in the natural image domain to the RS domain, and training domain-specific foundation models using large-scale synthetic datasets within the RS field. The direct transfer learning approaches enhance the performance of RSTIR by optimizing model architectures or training strategies [14], [15], [16]. Domain-specific foundation models, on the other hand, are developed using large-scale synthetic data generated based on specific rules or generative models [17], [18], [19]. Such methods are typically fine-tuned on large-scale synthetic datasets while maintaining their original pretrained architectures.

Despite recent advancements, mainstream approaches are suboptimal in fine-grained semantic alignment, struggling to capture fine-grained and semantically discriminative features. We identify two main reasons for this issue: the limited capability of previous state-of-the-art methods to extract and align fine-grained features, and the scarcity of datasets for fine-grained semantic alignment. Fig. 1 summarizes the two key motivations of this work. Specifically, the upper example illustrates that a query may contain several discriminative details,

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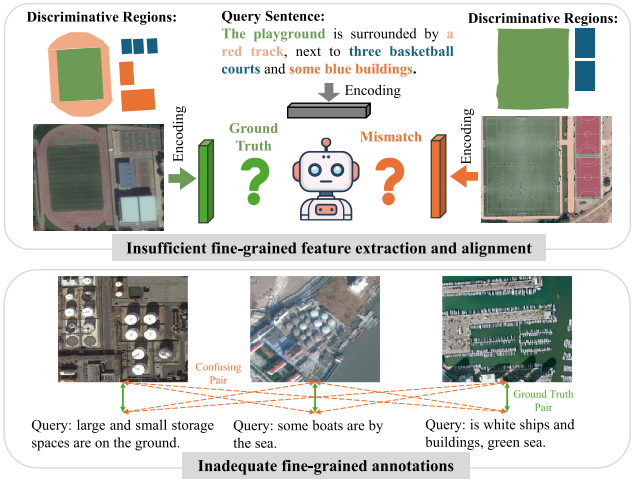


Fig. 1. Two challenges in fine-grained RS image–text retrieval. The upper example illustrates that global image–text matching may ignore discriminative local cues, such as three basketball courts and blue buildings, leading to fine-grained mismatches. The lower example shows that existing coarse and overlapping captions provide limited supervision for fine-grained retrieval. The green and red arrows, respectively, indicate the ground-truth pairs and confusing pairs.

such as the red track, three basketball courts, and blue buildings. However, when only global image–text embeddings are aligned, these local semantic cues may be insufficiently captured, leading to fine-grained mismatches. The lower example shows that several RSITMD captions contain highly overlapping and coarse descriptions. First, the insufficient extraction and alignment of fine-grained features is a key challenge in RSTIR. Traditional methods primarily encode the entire image and the whole sentence to obtain high-dimensional embedding vectors as visual and textual representations, respectively [20], [21]. Due to the lack of fine-grained visual representation, it is challenging to learn fine-grained visual-semantic correspondences. Most existing fine-grained RSTIR methods mainly depend on feature fusion or multitask learning, making it difficult to effectively align textual and visual semantics at a fine-grained level [14], [22]. To address this issue, we propose an innovative fine-grained text–image retrieval method (FTIR) for RSTIR. To effectively learn fine-grained semantically discriminative features from both RS images and queries, we incorporate region-level features [23] as supplementary cues. Specifically, inspired by the bottom-up attention mechanism [24], we first extract key regions from RS images, thus refining semantic granularity to the instance level. Subsequently, region-level features are aligned across modalities with word-level query representations. Meanwhile, a semantic importance gating mechanism is introduced to adaptively emphasize text segments that contribute most to the overall semantics. Moreover, in contrast to natural images, RS images are captured from a bird's-eye perspective. When transferring models that are well pretrained on the natural image domain to the RS domain, the model needs to cope with the large variations in object scales and semantic hierarchies. Therefore, direct fine-grained alignment in a high-dimensional common space tends to be less robust and easily affected by noise. To reduce the difficulty and noise sensitivity of fine-grained semantic alignment, we propose a multigranularity alignment strategy. It aligns both coarse-grained context information and fine-grained word-level information.

Second, as shown in Fig. 1, the lack of adequate fine-grained annotations is another challenge. Previous widely used datasets [22], [25] contain coarse-grained and highly repetitive textual queries. The overlapping coarse descriptions constrain the learning of detailed semantic distinctions. To facilitate fine-grained semantic alignment, we constructed a new dataset based on RSITMD. We have proposed RSITMD-FG, which comprises 23K image–text pairs with fine-grained semantic annotations. In contrast to traditional datasets annotated with short textual descriptions, our dataset assigns five fine-grained textual descriptions to each RS image. The average length of textual queries has increased from 11.3 to 32.2 words, offering richer semantic descriptions. Each textual query is carefully designed to emphasize distinct key entities. In addition to basic category labels, the attributes of key entities are enriched with information such as state, color, and quantity. Interentity relationships are also extended to include both spatial and functional interactions. Furthermore, each image is associated with up to 36 salient region features, enabling the model to capture spatially localized semantics effectively. Compared with traditional datasets, RSITMD-FG provides fine-grained annotations for both images and text.

To demonstrate FTIR's ability to capture fine-grained information and complex cross-modal correspondences, we evaluate on three benchmarks with varying levels of semantic granularity. Compared to previous approaches, our proposed framework achieves superior performance. Notably, our method surpasses some domain-specific foundation models without the need for additional training data or intricate cross-modal fusion mechanisms. Our main contributions are as follows.

1) We have proposed an innovative fine-grained framework for RSTIR. A region-level association module (RAM) is introduced to enhance the model's sensitivity to fine-grained semantics. A multigranularity alignment strategy is adopted to reduce the noise sensitivity of fine-grained semantic alignment.
2) We have constructed the RSITMD-FG dataset, which provides region-level image features and detailed textual queries. The RSITMD-FG is expected to facilitate further research in fine-grained RSTIR tasks.
3) The proposed FTIR achieves competitive performance on three public benchmarks, surpassing even some domain-specific foundation models. Code and data are publicly available to support reproducibility and future research.

II. Related Work

A. RS Text–Image Retrieval

The RSTIR task involves retrieving the most relevant text (or image) from a database given a query image (or text) [13], [26]. Deep learning-based image–text retrieval methods have gradually become the mainstream approaches owing to their powerful feature extraction capabilities. Depending on the data

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sources, current RSTIR methods can be broadly categorized into two types: transfer-based methods and synthetic data-based methods.

Given the scarcity of high-quality textual descriptions, transfer-based methods mainly focus on efficiently transferring the knowledge acquired in the natural image domain to the RS domain. These approaches directly utilize models pretrained on natural images, such as ResNet [27], CLIP [28], and GME [29]. Abdullah et al. [30] were the first to introduce the Deep Bidirectional Triplet Network for the RSTIR task and validated its effectiveness on the TextRS dataset. A multilanguage framework for RSTIR was proposed by Rahhal et al. [31], which enables broader access to RS information through multilanguage queries. Yuan et al. [22] developed an asymmetric multimodal feature matching model suitable for multisource inputs and proposed the RSITMD dataset. Methods such as CUP [21] and PE-RSITR [32] improve the efficiency of transfer learning by fine-tuning a subset of parameters.

Synthetic data-based methods use specific rules or generative models to produce large-scale synthetic data. These synthetic datasets are then used for supervised fine-tuning to improve model performance on RSTIR or scene classification tasks. Representative methods include RemoteCLIP [18], GeoRSCLIP [17], and Git-RSCLIP [19]. For tasks such as RSTIR, the RemoteCLIP and GeoRSCLIP methods expand the training data with 160K and 5M synthetic samples, respectively. For image generation and related tasks, Git-RSCLIP further extends the dataset with 10M synthetic samples. Muhtar et al. [33] proposed the ScoreRS model, constructed a dataset containing 72K image–text preference pairs, and subsequently evaluated its performance on the RSTIR task. Although some progress has been made, mainstream methods are suboptimal for achieving fine-grained semantic alignment. The lack of fine-grained information in existing RSTIR datasets constrains alignment accuracy, making it challenging to capture semantically discriminative fine-grained features. Therefore, enhancing the performance of fine-grained RSTIR requires refining the semantic granularity of both images and text while achieving accurate cross-modal alignment.

B. RS Image–Text Feature Alignment

Projecting the features of the two modalities into a shared semantic space while maintaining consistent semantic alignment between them remains one of the core challenges in RSTIR [34]. The inherent heterogeneity between visual and textual modalities often leads to a semantic gap, which hinders effective cross-modal retrieval [35]. To address this issue, recent studies have increasingly focused on exploring sophisticated semantic alignment strategies. They aim to construct accurate cross-modal relationships between modalities to enhance the consistency and interpretability of multimodal feature representations.

For instance, Pan et al. [36] leveraged prior knowledge from the RS domain to achieve adaptive alignment between visual and textual representations. GaLR [37] and GLISA [38] attempt to extract and align global and local features to enhance retrieval performance. Zhang et al. [39] exploited the difficulty levels of different triplet samples to design an adaptive alignment strategy progressing from easy to hard. Ji et al. [15] designed a novel eliminate-before-align strategy to filter out weakly correlated sample pairs, thereby mitigating their deviation from the optimal embedding space during the alignment process. Zheng et al. [40] proposed a cross-modal progressive perspective matching network for RSTIR by establishing a perspective matching mechanism and a semantic alignment strategy. Existing studies have explored different auxiliary alignment strategies in multimodal retrieval, including scene-aware supervision and keyword-level semantic guidance. Pan et al. [36] used scene category information as supplementary supervision to minimize the distance between the representation of each modality and the cluster center of its corresponding category. Yuan et al. [22] dynamically integrated sentence-level and keyword-level semantics. Ji et al. [15] proposed a keyword explicit reasoning module to leverage the beneficial role of differences among key concepts. However, these methods are usually designed for global matching or partial semantic enhancement. They do not explicitly address the challenges of fine-grained semantic modeling and alignment.

Although some methods have attempted to utilize fine-grained image representations, such as object-level feature extraction and fusion [37], they still do not explicitly establish the correspondence between individual words and specific image regions. Furthermore, for fine-grained alignment in a high-dimensional common space, previous strategies for globally aligning images and text are often not robust and are easily disturbed by noise. To address this issue, we introduce the novel network FTIR, which incorporates a RAM module and a multigranularity alignment strategy to achieve more precise fine-grained alignment.

III. Method

In this section, we present the proposed FTIR framework. Section III-A provides a brief introduction to the multimodal encoder. Section III-B provides a brief introduction to the RAM, while Section III-C details the multigranularity alignment strategy. The pipeline of our proposed FTIR is shown in Fig. 2.

A. Multimodal Encoder

For the multimodal encoder, we follow mainstream approaches by employing the ViT-based model [41] to extract visual features and a text transformer model [42] for textual features. The extracted visual and textual features are projected into a common embedding space [39]. The similarity between visual and textual embeddings is measured by the normalized inner product. The above process can be expressed mathematically as

$$f_v = \varphi_v(I) \in \mathbb{R}^{N_v \times h_v} \tag{1}$$

$$f_v = \left[f_v^{cls}, f_v^0, f_v^1, \ldots, f_v^{N_v-1}\right] \tag{2}$$

where I denotes the input RS image, and $\varphi_v(\cdot)$ represents the visual encoder, which extracts visual features f_v using ViT. The RS image I is partitioned into nonoverlapping patches of

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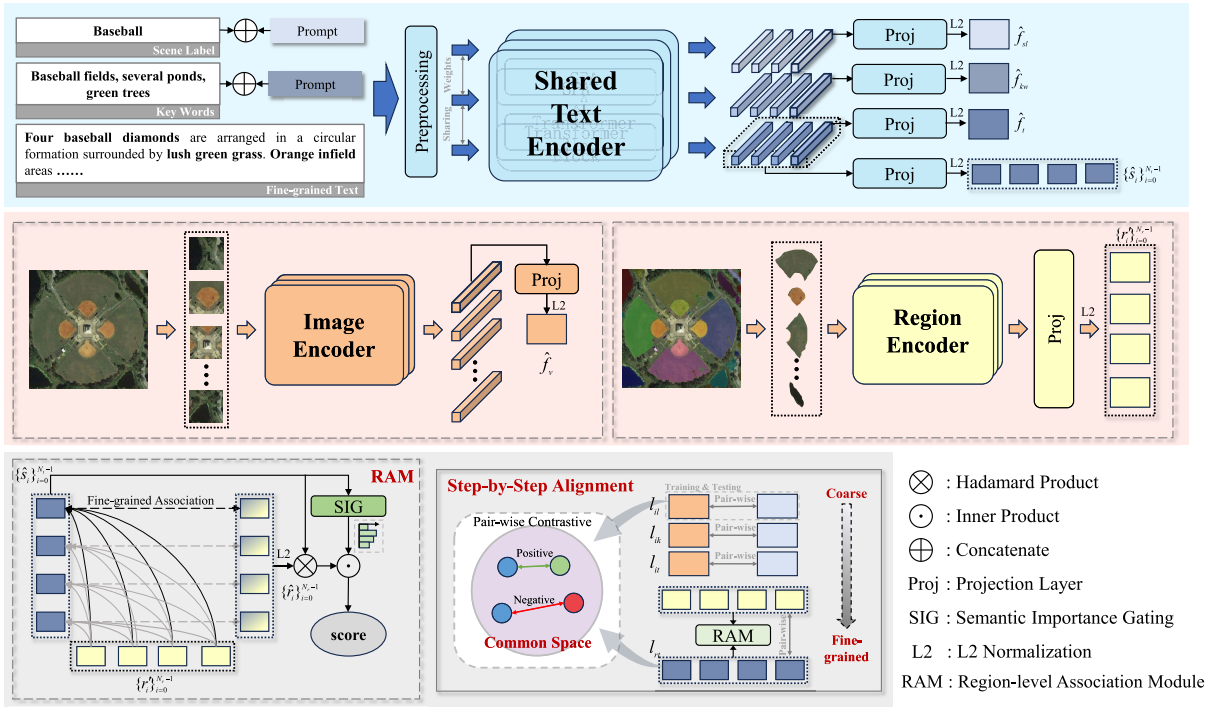


Fig. 2. Pipeline of the proposed FTIR method. Semantic information at different granularities is extracted by the shared text encoder. The global and regional image features are obtained by the image encoder and the region encoder, respectively. Fine-grained associations between region features and word-level features are established via the RAM module. During alignment, a multigranularity alignment is employed to align features with both coarse and fine granularity.

fixed size, yielding N_v patches in total, where h_v denotes the dimensionality of each patch. An additional [CLS] token is appended to the input sequence, and the output corresponding to this token from the final ViT block, denoted as f_v^{cls}, is used as the global visual representation of the entire image. The final visual embedding $\hat{f}_v$ is obtained through projection and normalization

$$\hat{f}_v = \left\| proj\left(f_v^{cls}\right) \right\|_2 \tag{3}$$

where $proj(\cdot)$ represents the projection function, and $\|\cdot\|_2$ represents L2 normalization.

Similarly, the process of extracting textual features can be formulated as follows:

$$f_t = \varphi_t(T) \in \mathbb{R}^{N_t \times h_t} \tag{4}$$

$$f_t = \left[f_t^{bos}, f_t^0, f_t^1, \ldots, f_t^{N_t-1}, f_t^{eos} \right] \tag{5}$$

where T denotes the textual query, and $\varphi_t(\cdot)$ represents the text encoder, which extracts textual features f_t using a Transformer-based architecture. The input text T is tokenized into a sequence of N_t tokens, with h_t being the dimensionality of each token. An [EOS] token is appended to the token sequence, and the output corresponding to this token from the final Transformer block, denoted as f_t^{eos}, serves as the global representation of the entire textual query.

B. Region-Level Association Module

Traditional approaches mainly perform contrastive learning at the global level, aligning entire image representations with whole textual queries. However, such coarse alignment overlooks the fine-grained correspondences that are crucial for RS images. Although several methods have explored fine-grained image feature extraction, such as object-level representations, they do not explicitly establish the correspondence between words and image regions. These methods fail to refine the semantic granularity of the text modality, resulting in suboptimal performance when processing fine-grained semantic queries. Therefore, achieving simultaneous refinement and alignment of semantic granularity across both image and text modalities is crucial for effective fine-grained image–text retrieval in the RS domain.

To overcome these limitations, we introduce a RAM, which enables fine-grained alignment between visual regions and textual segments in a lightweight and flexible manner. Inspired by the bottom-up attention mechanism [24], [43], [44], we aim to establish associations between salient regions in the image and important words in the textual query. The details of the RAM are shown in Fig. 3.

Specifically, our approach proceeds as follows. Initially, we employ an off-the-shelf method, Segment Anything Model 2 (SAM2), to segment the RS image into semantically significant units, reflecting human perceptual intuition [45], [46]. After pooling, the regional feature R_i is obtained. Next, instead of representing the entire query as a single vector, we extract individual features for each word

$$r = \{r_i\}_{i=0}^{N_r-1} = \varphi_r\left(\{R_i\}_{i=0}^{N_r-1}\right) \tag{6}$$

$$s = \{s_i\}_{i=0}^{N_t-1} = \phi\left(\{f_t^i\}_{i=0}^{N_t-1}\right) \tag{7}$$

where R_i denotes the ith regional feature in the image, $\varphi_r(\cdot)$ represents the region used to extract the final region-level features r, and N_r is the total number of regions. The region

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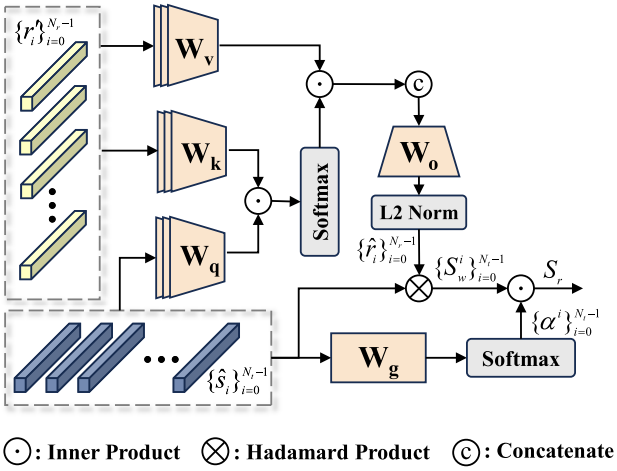


Fig. 3. Application details of the proposed RAM. The textual segments serve as queries. The region features serve as keys and values. The region-aware feature is obtained via cross-modal attention. The final similarity score is derived by applying a semantic importance gating mechanism to the region-aware feature and textual segments.

encoder adopts the BERT-base architecture. The function $\phi(\cdot)$ denotes the refinement module that refines the query features to textual segments s.

To enable fine-grained semantic alignment, we establish cross-interactions between each textual segment and region feature. Each region-aware feature is obtained by aggregating information from multiple region features conditioned on the specific textual segment

$$H_j = \text{softmax}\left(\frac{W_j^q s\left(W_j^k r\right)^\top}{\sqrt{d_h}}\right) W_j^v r \tag{8}$$

$$\left\{r_i'\right\}_{i=0}^{N_t-1} = \left\{H_j\right\}_{j=0}^{h-1} W^o \tag{9}$$

where H_j denotes the jth attention head. W_j^q, W_j^k, W_j^v, and W^o are learnable weight matrices. r_i' represents the ith region-aware feature with a total number of N_r. The obtained region-aware features and textual segment features are projected into a shared embedding space to perform semantic alignment at the region level

$$\hat{r}_i = \|proj\left(r'_i\right)\|_2 \tag{10}$$

$$\hat{s}_i = \|proj\left(s_i\right)\|_2 \tag{11}$$

$$S_w^i = \langle \hat{r}_i, \hat{s}_i \rangle \tag{12}$$

where $proj(\cdot)$ represents the projection function, and S_w^i represents the similarity between ith region-aware feature and ith textual segment feature. $\langle\cdot\rangle$ denotes the inner product. $\|\cdot\|_2$ represents L2 normalization.

Considering that different textual segments contribute unequally, we propose a semantic importance gating mechanism. The semantic importance gating mechanism is designed to emphasize significant words within the textual queries, such as nouns and numerals, while suppressing less informative words, such as conjunctions. The final region-level semantic similarity can be computed as

$$\left\{\alpha^i\right\}_{i=0}^{N_t-1} = \text{softmax}\left(W_g\left\{\hat{s}_i\right\}_{i=0}^{N_t-1}\right) \tag{13}$$

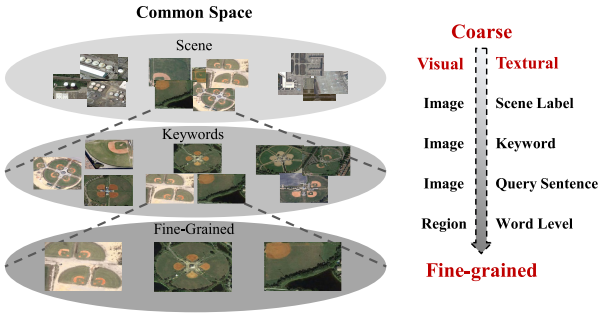


Fig. 4. MSA. Cross-modal alignment is achieved across four hierarchical levels: scene-level, keyword-level, sentence-level, and word-level.

$$S_r = \sum_{i=0}^{N_t-1} \alpha^i S_w^i \tag{14}$$

where W_g denotes the gating matrix, α represents the gating weights, S_w^i represents the similarity between ith region-aware feature and ith textual segment feature, and S_r denotes the final region-level semantic similarity score.

C. Multigranularity Semantic Alignment

Compared with natural images, RS images are captured from a bird's-eye perspective, resulting in a unique viewpoint. RS images exhibit significant variations in object scales and semantic hierarchies, which is manifested as high variance in image entropy [39], [47]. When directly transferring models well pretrained on natural image domains to the RS domain for fine-grained semantic alignment, the diverse object scales and complex spatial distributions may make the model susceptible to noise and difficult to converge during training. To address this issue, as shown in Fig. 4, we propose a multigranularity semantic alignment (MSA) strategy. This approach enables the model to jointly optimize across hierarchical levels.

Unlike previous state-of-the-art methods [40], [48], we propose to perform semantic alignment from multiple levels of granularity. Specifically, we define four hierarchical levels, ranging from coarse to fine: scene-level, keyword-level, sentence-level, and word-level. To unify the optimization of different alignment stages, we follow the approach of SigLIP [49], [50] by formulating the semantic alignment process as a binary classification task. For a mini-batch $\mathcal{B}$ of size $|\mathcal{B}|$, each sample $\hat{x}_i \in X$ is paired with each sample $\hat{y}_j \in Y$, resulting in $|\mathcal{B}|^2$ cross-modal pairs. Here, $\hat{x}_i$ and $\hat{y}_j$ are the projected and normalized embeddings in the shared semantic space. The binary label $Z_{i,j}$ indicates whether the pair is semantically matched, where $Z_{i,j} = 1$ for a positive pair and $Z_{i,j} = 0$ for a negative pair

$$\mathcal{L}(X,Y) = -\frac{1}{|\mathcal{B}|^2}\sum_{i=1}^{|\mathcal{B}|}\sum_{j=1}^{|\mathcal{B}|}\left[Z_{i,j}\cdot\log\left(P\left(\hat{x}_i,\hat{y}_j\right)\right) + \left(1-Z_{i,j}\right)\cdot\log\left(1-P\left(\hat{x}_i,\hat{y}_j\right)\right)\right] \tag{15}$$

$$P\left(\hat{x}_i,\hat{y}_j\right) = \sigma\left(\tau\langle\hat{x}_i,\hat{y}_j\rangle - b\right) = \frac{1}{1+e^{-\left(\tau\langle\hat{x}_i,\hat{y}_j\rangle-b\right)}} \tag{16}$$

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where X and Y denote the representations from two modalities to be aligned, and $\mathcal{L}(X, Y)$ denotes the corresponding cross-modal alignment loss. $P(\hat{x}_i, \hat{y}_j)$ represents the predicted probability that the pair is matched. Therefore, (15) can be interpreted as averaging the binary cross-entropy over all cross-modal pairs in the mini-batch, encouraging matched pairs to obtain high similarity scores and unmatched pairs to obtain low similarity scores. In (16), $\langle \hat{x}_i, \hat{y}_j \rangle$ denotes the inner product between $\hat{x}_i$ and $\hat{y}_j$, which measures their similarity. τ is a scaling factor that controls the sharpness of the similarity score, b is a learnable bias term, and $\sigma(\cdot)$ denotes the sigmoid function.

At the scene stage, our goal is to cluster RS images with the same scene label in the common embedding space. We constrain the alignment between images and textual queries that include the same scene label. To construct these queries, we design prompts such as "the scene label is [SCENE_LABEL]" to explicitly incorporate scene information

$$l_{il} = \mathcal{L}(I, Q_{label}) \tag{17}$$

where Q_{label} represents the constructed textual queries with RS scene label, and $\mathcal{L}$ denotes the semantic alignment loss between the image I and label-based query Q_{label}.

In contrast to the scene-level stage, the objective of the keyword-level alignment is to enable the model to learn correspondence between specific keywords and RS images. For samples lacking keywords, we extract key phrases from the original textual queries using part-of-speech tagging [51]. We extract adjective-noun phrases, numeral-noun phrases, and other semantically informative constructs from the query text as keywords

$$Q_{keyword} = \{q_i \in T | \phi_{pos}(q_i) \in \mathcal{T}\} \tag{18}$$

$$l_{ik} = \mathcal{L}(I, Q_{keyword}) \tag{19}$$

where q_i denotes the ith word in the textual query, and $\phi_{pos}(q_i)$ represents the part-of-speech tagging function applied to q_i. $\mathcal{T}$ denotes the predefined set of keyword-related part-of-speech tags, and $Q_{keyword}$ is the final query composed of the extracted keywords. l_{ik} denotes the semantic alignment loss between the RS image I and the keyword-based query.

Based on the aforementioned descriptions, the semantic alignment losses at the sentence level and word level are denoted as l_{it} and l_{rw}, respectively. The sentence-level alignment follows the same strategy as prior mainstream methods [17], in which the representation of the entire query is aligned with that of the whole image. Word-level alignment refers to the fine-grained alignment between image regions and textual segments, which is achieved through the previously proposed RAM. Specifically, RAM first computes the region-aware word-level similarity according to (8)–(14). We then use S_r as the matching score in the pairwise binary alignment objective defined in (15) and (16), and denote the resulting loss as l_{rw}

$$l_{rw} = \mathcal{L}(r, s) \tag{20}$$

$$P(r, s) = \sigma(\tau S_r - b) \tag{21}$$

where r and s correspond to the region-level features and textual segments in (6) and (7), respectively. S_r denotes the final similarity score calculated using (14). $P(r, s)$ represents the matching probability.

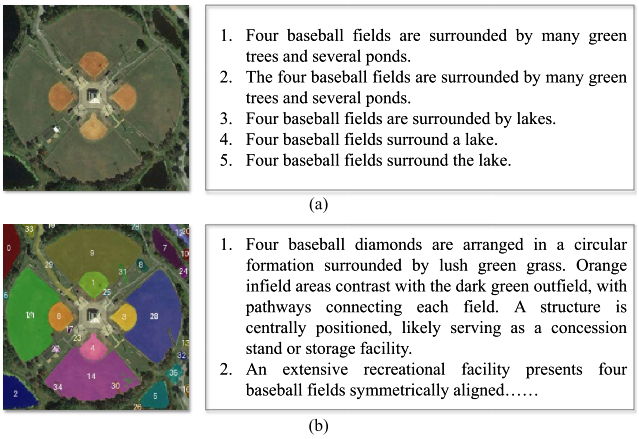


Fig. 5. Comparison of samples from different datasets: (a) original RSITMD dataset and (b) RSITMD-FG dataset. Important regions in the image are marked using masks and numbers.

The overall training objective of our proposed multigranularity alignment is defined as minimizing the total loss

$$l_{total} = e^{-\lambda_0} l_{il} + e^{-\lambda_1} l_{ik} + l_{it} + e^{-\lambda_2} l_{rw}. \tag{22}$$

The sentence-level alignment l_{it} serves as the core task of our cross-modal retrieval framework, providing the global semantic anchor for the joint embedding space. λ_0, λ_1, and λ_2 are the scaling factors to ensure that other loss components are optimized on a comparable magnitude at the beginning of training.

Although scene-level alignment and keyword-level supervision have been explored in previous studies, existing methods still lack a unified framework that can explicitly achieve alignment across different semantic granularity levels. Unlike methods that introduce auxiliary alignment supervision separately, we emphasize the joint integration of these different levels of supervision to enhance fine-grained semantic alignment.

IV. Dataset

The RSICD dataset contains 10 921 RS images collected from Google Earth, covering 30 scene categories [25]. Each image, with a size of 224 × 224 pixels and varying resolution, is annotated with five textual descriptions. The average length of query texts in RSICD is 11.5 words. However, the textual descriptions in RSICD are relatively coarse-grained and exhibit high redundancy. To address these limitations, Yuan et al. [22] introduced the RSITMD dataset, which consists of 4743 RS images sourced from RSICD and Google Earth across 32 categories. Each image has a resolution of 256×256 pixels and is associated with five textual descriptions that provide more diverse semantic information compared with those in RSICD, as illustrated in Fig. 5(a). The average length of query texts in RSITMD is 11.3 words.

However, both RSICD and RSITMD lack fine-grained annotations. The absence of fine-grained annotations is one of the key factors limiting the performance of previous mainstream

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methods, leading to confusion among visually or textually similar samples. To advance the development of fine-grained RSTIR, we propose a new benchmark named RSITMD-FG. As illustrated in Fig. 5(b), RSITMD-FG is an updated version of the RSITMD, which we previously introduced. We enhanced fine-grained semantic annotations for both the image and query text in RSITMD-FG. Compared with existing datasets, the captions in our dataset are longer and semantically richer. The average caption length has increased from 11.3 to 32.2 words, providing more detailed descriptions of RS images. Each image is annotated with up to 36 salient region features, enabling the model to capture spatially localized semantics more effectively.

The construction of the RSITMD-FG dataset comprises four steps.

1) *Fine-Grained Region-Level Annotation of RS Images:* First, we employ the SAM2 [46] to extract salient region features from RS images. For each RS image, we uniformly sample 64×64 points and use SAM2 to obtain the corresponding region features for each point after pooling. The top 36 regions with the highest confidence scores are then selected as the fine-grained features of the image. In our opinion, 36 region features are sufficient to represent the key geographical entities within an image, which is consistent with prior related research [24], [37], [44].
2) *Fine-Grained Text Generation Using Large Multimodal Models:* For each image, we employ GPT-4o [52] to generate the initial fine-grained textual descriptions, where scene categories, keywords, and annotated coarse-grained texts are incorporated into the prompt as prior knowledge. Five diverse fine-grained query texts are generated for each image. Each query is carefully constructed to emphasize distinct key entities, ensuring diversity and descriptive depth. In the prompt, we explicitly defined the criteria for fine-grained semantic enhancement. Beyond basic category information, the textual annotations incorporate additional attributes, such as state, color, and quantity, to achieve finer semantic granularity. Furthermore, interentity relationships are enriched to capture not only spatial arrangements but also functional interactions among objects. The maximum length of each text description is limited to 50 words.
3) *Refinement of Textual Queries to Enhance Diversity and Relevance:* We re-evaluate the fine-grained textual descriptions using GPT4o for image–text relevance judgment, and refine the inaccurate fine-grained textual descriptions.
4) *Manual Verification of All Data Samples to Ensure Overall Quality and Consistency:* Five annotators review and revise the final generated fine-grained textual descriptions.

To ensure annotation quality, all generated descriptions are manually reviewed and revised before being included in RSITMD-FG. The verification process consists of two aspects.

1) *Entity-Attribute Level:* For each entity, we examine whether the generated description is accurate with respect to four fine-grained attributes, namely category, state, color, and quantity.
2) *Interentity Relationship Level:* We further verify whether the generated text accurately captures the spatial and functional relationships among multiple entities. Here, spatial relationships mainly refer to the relative positions or distribution patterns among entities, whereas functional relationships refer to the associations in their practical uses or roles within real-world scenes. Five annotators check and correct the generated results to ensure data reliability and sufficient semantic granularity. Through this human-in-the-loop process, RSITMD-FG aims to provide more accurate and fine-grained supervision for training.

TABLE I
COMPARISON WITH PREVIOUS RS VISION-LANGUAGE DATASETS

Dataset	Image		Textual Query		Human
	Num	Fine-Grained	Num	Fine-Grained	
Sydney [53]	555	✗	2,775	✗	✓
UCM-Captions [53]	2,100	✗	10,500	✗	✓
RSICD [25]	10,921	✗	54,605	✗	✓
RSITMD [22]	4,743	✗	23,715	✓✗	✓
RSICap [54]	2,585	✗	2,585	✓	✓✗
RS5M [17]	5M	✗	5M	✓	✗
VRSBench [55]	29,614	✓✗	29,614	✓	✓
XLRSBench [56]	934	✓✗	934	✓	✗
RSITMD-FG	4,743	✓	23,715	✓	✓

A comparative analysis of RSITMD-FG and existing benchmarks is presented in Table I. The RSITMD-FG dataset comprises 4743 RS image patches with a spatial resolution of 256×256 pixels. Each image is annotated with 36 region-level features representing key semantic areas. Each image is also annotated with five fine-grained textual queries, yielding a total of 23 715 image–text pairs. Compared with conventional datasets, such as UCM-Captions [53], RSITMD [22], and RSICD [25], RSITMD-FG provides finer semantic granularity and stronger quality. When compared with the large language model-generated RS5M [17], our RSITMD-FG provides more reliable annotations. Our dataset has been fully verified manually. In contrast to more recent benchmarks such as VRSBench [55] and XLRS-Bench [56], our dataset focuses specifically on the retrieval of RS image patches and incorporates more detailed region-level features.

V. EXPERIMENTS

A. Experimental Setup

Following previous advanced methods, FTIR is initialized with the weights of pretrained backbones and evaluated on representative benchmark datasets using the same data splits for fair comparison. All experiments are conducted using PyTorch [57] on two NVIDIA A20 GPUs. Following the same data split as previous studies [22], 80%, 10%, and 10% of the samples from the RSICD, RSITMD, and RSITMD-FG datasets are used for training, validation, and testing, respectively. In FTIR, the visual and textual encoders are initialized with pretrained weights from SigLIP2 [50], whereas the region encoder is initialized with the pretrained weights of BERT-base. All RS images are resized to 384×384 or

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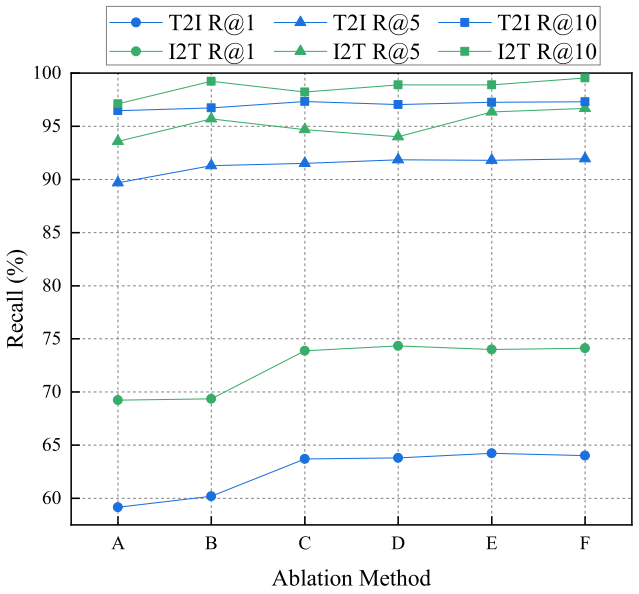


Fig. 6. Detailed comparison of different ablation methods on the RSITMD-FG dataset. T2I denotes the text-to-image retrieval. I2T denotes the image-to-text retrieval task.

TABLE II
ABLATION STUDIES ABOUT FTIR

Method	RAM	Scene-Level	Keyword-Level	Loss	RSITMD-FG	RSICD
F	✓	✓	✓	Paired	87.30	43.10
E	✓	✓		Paired	87.09	42.93
D	✓		✓	Paired	86.65	42.84
C	✓			Paired	86.56	42.25
B				Paired	85.42	41.65
A				Triplet	84.21	39.70

256 × 256 pixels, and the maximum text length is set to 64. We evaluated the performance of FTIR using different visual backbone networks, including ViT-B-16 and ViT-B-32. Unless otherwise specified in the experiments, FTIR adopts ViT-B-16 as the default visual backbone.

Recall at K (R@K) is adopted to assess retrieval performance at different thresholds, while mean recall (mR) reflects the overall average retrieval effectiveness. We use AdamW [58] as the optimizer with an initial learning rate of 10^{-5} and a weight decay of 0.05. The values of λ_0, λ_1, and λ_2 are all set to 3. The batch size is set to 30, and the model is trained for a total of 20 epochs. Additional implementation details are provided in our open-source repository.

B. Ablation Studies

To evaluate the effectiveness of the key components in FTIR, we have conducted ablation studies on both the dataset with fine-grained annotations (RSITMD-FG) and the dataset without fine-grained annotations (RSICD). Specifically, we individually remove the RAM module, semantic alignment at different levels, and various semantic alignment loss functions. Table II reports the mR metric of various ablation experiments on RSICD and RSITMD-FG. Fig. 6 illustrates the impact of ablation studies in detailed recall metrics on the RSITMD-FG.

The comparison of methods D, E, and F on the RSITMD-FG dataset demonstrates that scene-level semantic alignment and keyword-level semantic alignment can both effectively enhance retrieval performance. The scene-level semantic alignment is more effective, as its removal resulted in a 0.62% decrease in overall retrieval performance. The comparison between methods C and F shows the advantage of the proposed multigranularity alignment strategy. The traditional one-step alignment lags behind by 0.71% in the mR metric. The comparison between methods B and C shows that removing the RAM leads to a 1.14% drop. This result highlights the importance of region-level association for fine-grained RSTIR retrieval. In the comparison between methods A and B, we find that using conventional triplet-based semantic alignment loss functions has slightly inferior results.

TABLE III
SENSITIVITY ANALYSIS OF THE PARAMETERS $\lambda_{0,1,2}$ ON THE RSITMD-FG DATASET

λ_0	λ_1	λ_2	Image to Text R@1	Image to Text R@5	Text to Image R@1	Text to Image R@5	mR
0	0	0	69.69	94.91	60.61	90.57	85.16
1	1	1	75.44	95.57	63.84	91.28	87.10
3	3	3	76.54	96.23	63.93	91.54	87.30
5	5	5	73.23	95.13	62.38	91.19	86.32

We further present detailed changes in various evaluation metrics, as illustrated in Fig. 6. Comparing methods B and C, we find that the introduction of the RAM module improves the R@1 significantly for both text-to-image retrieval and image-to-text retrieval. This demonstrates that the proposed RAM effectively addresses the limitations of traditional methods in capturing fine-grained semantic information. The incorporation of the multigranularity alignment strategy has limited improvement in R@1, but leads to noticeable gains on less strict metrics such as R@5 and R@10. Our proposed RAM and multigranularity alignment strategy complement each other, enabling the FTIR framework to achieve competitive performance.

On the RSICD dataset, a key observation is that the RAM module provides limited performance improvement due to the coarse semantic granularity of the dataset. Compared to RAM, the multigranularity alignment strategy obtains a more significant performance gain (+0.85%). In addition, the triplet loss function results in a noticeable performance degradation on the RSICD dataset. One reason is that the inconsistency with the optimization objective of the pretrained model reduced the performance of transfer learning. Another reason is that the high similarity of query texts in RSICD generated more false negative samples that interfered with the training process.

To investigate the impact of the scaling parameters $\lambda_{0,1,2}$ on the retrieval performance, we conducted a sensitivity analysis on the RSITMD-FG dataset. As illustrated in Table III, we varied the values of scaling parameters $\lambda_{0,1,2}$ from 0 to 5 while keeping the primary sentence-level loss weight fixed.

During the initial phase of training, we observed that the raw values of l_{il}, l_{ik}, and l_{rw} are significantly larger than the primary sentence-level loss l_{it}. Since the pretrained backbone has not been optimized with the proposed scene-level, keyword-level, and word-level auxiliary supervision, the raw values of l_{il}, l_{ik},

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TABLE IV
COMPARISON WITH THE STATE-OF-THE-ART MODELS ON RSICD, RSITMD

Approach	Visual Encoder	Train Set	Test Set	Image to Text			Text to Image			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
$AMFMN_{TGRS2022}$ [22]	ResNet-18	RSICD	RSICD	5.39	15.08	23.40	4.90	18.28	31.44	16.42
$KCR_{IJCAI2022}$ [26]	ResNet-101	RSICD	RSICD	5.95	19.59	29.58	5.40	22.44	37.36	19.89
$Multilanguage_{JSTARS2022}$ [31]	ViT-B-32	RSICD	RSICD	10.70	29.64	41.53	9.14	28.96	44.59	27.42
$SWAN_{ICMR2023}$ [59]	ResNet-50	RSICD	RSICD	7.41	20.13	30.86	5.56	22.26	37.41	20.61
$FAMMI_{TGRS2023}$ [60]	DetNet	RSICD	RSICD	10.44	22.66	30.89	8.11	25.59	41.37	23.18
$PIR_{ACMMM2023}$ [36]	Swin-T	RSICD	RSICD	9.88	27.26	39.16	6.97	24.56	38.92	24.46
$KAMCL_{TGRS2023}$ [61]	ResNet-101	RSICD	RSICD	12.08	27.26	38.70	8.65	27.43	42.51	26.10
$PE\text{-}RSITR_{TGRS2023}$ [32]	ViT-B-32	RSICD	RSICD	14.13	31.51	44.78	11.63	33.92	50.73	31.12
$RemoteCLIP_{TGRS2024}$ [18]	ViT-B-32	RET-3+DET-10+SEG-4(0.82M)	RSICD	17.02	37.97	51.51	13.71	37.11	54.25	35.26
$RemoteCLIP_{TGRS2024}$ [18]	ViT-L-14	RET-3+DET-10+SEG-4(0.82M)	RSICD	18.39	37.42	51.05	14.73	39.93	56.58	36.35
$GeoRSCLIP_{TGRS2024}$ [17]	ViT-B-32	RS5M+RET-2(5M+0.07M)	RSICD	21.13	41.72	55.63	15.59	41.19	57.99	38.87
$EBAKER_{ACMMM2024}$ [15]	ViT-B-32	RSICD+RSITMD+NWPU(0.2M)	RSICD	21.87	44.46	58.92	17.37	43.00	58.55	40.70
$CLGSA_{TGRS2025}$ [48]	ViT-B-16	RSICD	RSICD	14.38	36.73	45.35	11.75	34.07	50.12	32.07
$CPPMN_{TMM2025}$ [40]	ViT-L-14	RSICD	RSICD	19.12	39.57	53.89	15.61	41.22	56.96	37.72
FTIR(OURS)	ViT-B-32	RSICD	RSICD	21.77	44.54	58.91	17.63	43.92	59.56	41.06
FTIR(OURS)	ViT-B-16	RSICD	RSICD	**25.25**	**45.92**	**59.01**	**20.49**	**46.66**	**61.28**	**43.10**
$AMFMN_{TGRS2022}$ [22]	ResNet-18	RSITMD	RSITMD	11.06	29.20	38.72	9.96	34.03	52.96	29.32
$Multilanguage_{JSTARS2022}$ [31]	ViT-B-32	RSITMD	RSITMD	19.69	40.26	54.42	17.61	49.73	66.59	41.38
$SWAN_{ICMR2023}$ [59]	ResNet-50	RSITMD	RSITMD	13.35	32.15	46.90	11.24	40.40	60.60	34.11
$FAMMI_{TGRS2023}$ [60]	DetNet	RSITMD	RSITMD	16.15	35.62	48.89	12.96	42.39	59.96	35.99
$PIR_{ACMMM2023}$ [36]	Swin-T	RSITMD	RSITMD	18.14	41.15	52.88	12.17	41.68	63.41	38.24
$KAMCL_{TGRS2023}$ [61]	ResNet-101	RSITMD	RSITMD	16.51	36.28	49.12	13.50	42.15	59.32	36.14
$PE\text{-}RSITR_{TGRS2023}$ [32]	ViT-B-32	RSITMD	RSITMD	23.67	44.07	60.36	20.10	50.63	67.97	44.47
$RemoteCLIP_{TGRS2024}$ [18]	ViT-B-32	RET-3+DET-10+SEG-4(0.82M)	RSITMD	27.88	50.66	65.71	22.17	56.46	73.41	49.38
$RemoteCLIP_{TGRS2024}$ [18]	ViT-L-14	RET-3+DET-10+SEG-4(0.82M)	RSITMD	28.76	52.43	63.94	23.76	59.51	74.73	50.52
$GeoRSCLIP_{TGRS2024}$ [17]	ViT-B-32	RS5M+RET-2(5M+0.07M)	RSITMD	32.30	53.32	67.92	25.04	57.88	74.38	51.81
$EBAKER_{ACMMM2024}$ [15]	ViT-B-32	RSICD+RSITMD+NWPU(0.2M)	RSITMD	**34.07**	54.20	**67.95**	28.05	**60.35**	**75.31**	53.32
$CLGSA_{TGRS2025}$ [48]	ViT-B-16	RSITMD	RSITMD	26.77	48.23	60.39	22.52	52.39	69.46	46.63
FTIR(OURS)	ViT-B-32	RSITMD	RSITMD	31.41	55.64	67.81	27.99	57.59	72.06	52.08
FTIR(OURS)	ViT-B-16	RSITMD	RSITMD	32.30	**55.97**	67.47	**29.07**	**60.35**	74.73	**53.32**

TABLE V
COMPARISON WITH THE STATE-OF-THE-ART MODELS ON RSITMD-FG

Approach	Visual Encoder	Image to Text			Text to Image			mR
		R@1	R@5	R@10	R@1	R@5	R@10	
AMFMN [22]	ResNet-18	24.33	59.60	75.78	19.44	48.17	64.58	48.65
EBAKER [15]	ViT-B-32	36.72	76.99	89.38	25.66	63.67	81.23	62.27
RemoteCLIP [18]	ViT-B-32	36.72	77.43	89.82	29.82	69.33	86.28	64.90
RemoteCLIP [18]	ViT-L-14	50.57	86.85	94.25	38.40	79.29	93.09	73.74
CLIP [28]	ViT-B-16	44.02	81.19	90.93	33.01	73.14	87.78	68.35
BLIP [62]	ViT-B-16	63.71	93.36	97.78	56.32	88.93	96.10	82.70
SIGLIP [49]	ViT-B-16	67.69	94.69	98.23	60.39	90.08	96.01	84.52
BLIP2 [63]	ViT-L-14	65.48	91.59	96.68	58.71	91.15	**97.21**	83.47
SIGLIP2 [50]	ViT-B-16	73.23	94.69	**98.67**	61.85	91.06	96.90	86.07
FTIR(OURS)	ViT-B-32	72.12	92.92	96.90	61.54	90.88	96.72	85.18
FTIR(OURS)	ViT-B-16	**76.54**	**96.23**	98.45	**63.93**	**91.54**	97.12	**87.30**

and l_{rw} are relatively large at the beginning of training. To prevent the gradients of these auxiliary tasks from overwhelming the primary optimization direction, the scaling parameters $\lambda_{0,1,2}$ should be greater than 0. Therefore, we set $\lambda_{0,1,2}$ to the same value mainly to keep the scaled auxiliary losses at a comparable numerical level to l_{it} at the start of training. This shared setting is used for numerical scale balancing. It also reduces the hyperparameter search space and improves the simplicity and reproducibility of the framework. As observed, when $\lambda = 0$, all loss components are weighted equally. In this case, the mR is only 85.16. This relatively poor performance is due to gradient imbalance. This phenomenon supports our initial motivation: excessively large weights for auxiliary tasks lead to gradient dominance, where the model over-optimizes for local entities or scene labels at the expense of the global sentence-image semantic consistency. Conversely, when $\lambda_{0,1,2}$ is set to a large value 5, both $R@1$ and $R@5$ begin to decline. The performance gain from multigranularity alignment is limited. The peak performance at $\lambda_{0,1,2} = 3$ confirms that maintaining a comparable gradient magnitude between the sentence-level and auxiliary levels is crucial for effective joint learning.

C. Comparison With the State-of-the-Art Models

We have compared our approach with previous state-of-the-art methods across datasets with varying levels of semantic granularity, summarized in Tables IV and V.

1) Performance on the RSICD: Our proposed method obviously outperforms other similar transfer-based methods

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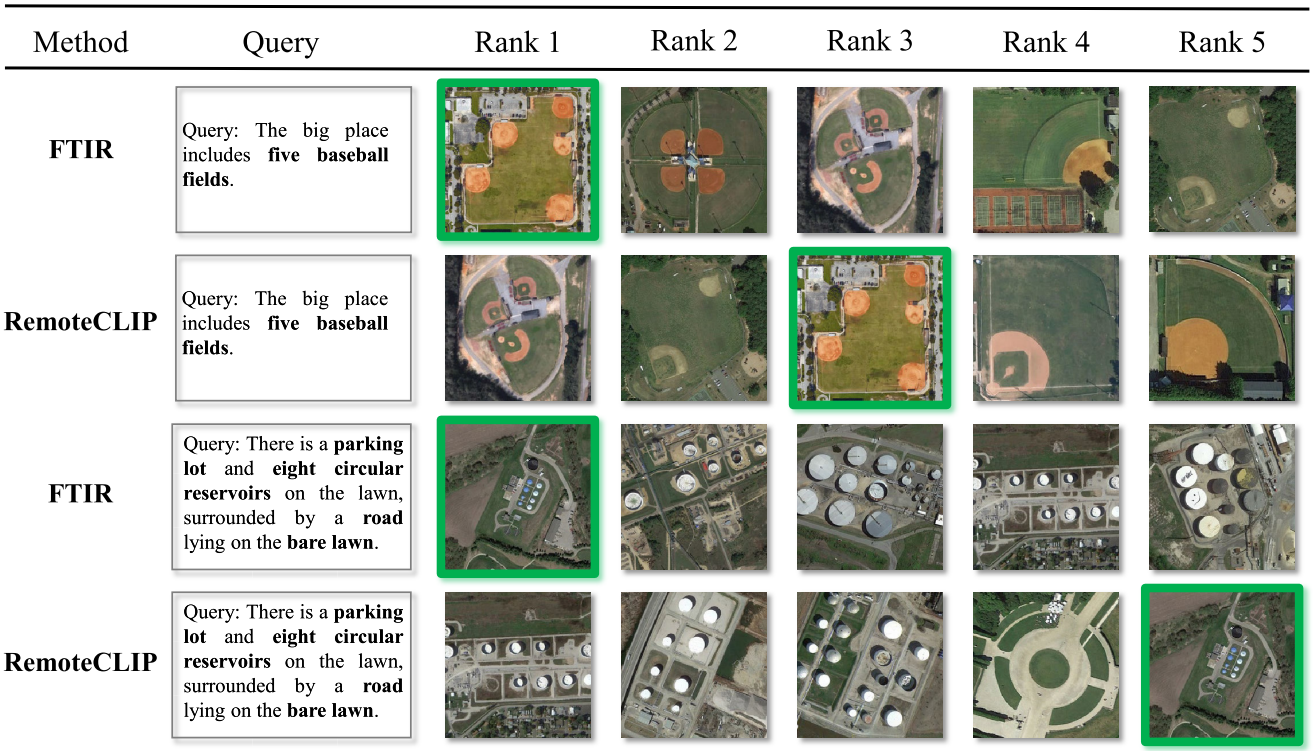


Fig. 7. Comparison of text-to-image retrieval results on the RSITMD dataset using different methods. Key information in the query text is highlighted in bold. Images within the green boxes represent the ground truth. The retrieved images are ranked in descending order of similarity.

when using the same training set. FTIR achieves an 18.64% improvement in the mR metric over the PIR method [36]. Compared to the CLGSA method [48], our proposed method achieves an improvement of 8.74% in the R@1 metric on the text-to-image retrieval task. Compared to the EBAKER method [15], our method achieves a better average retrieval performance (+2.40%) with less training data. FTIR with ViT-B-32 achieves an mR of 41.06, outperforming EBAKER with the same backbone. Compared to methods based on synthetic data, such as RemoteCLIP [18] and GeoRSCLIP [17], our proposed method also outperforms these RS-specific models. Notably, this performance improvement is achieved without incorporating large amounts of additional training data, which highlights the effectiveness and efficiency of the proposed FTIR method.

2) Performance on the RSITMD: Our FTIR framework achieves superior performance compared with other transfer-based approaches. Although EBAKER [15] achieves performance comparable to FTIR, our proposed FTIR requires only a quarter of the training data. Our method still outperforms EBAKER on the text-to-image retrieval task, achieving a 1.02% improvement in the R@1 metric. FTIR is indeed slightly lower than EBAKER on several recall metrics. FTIR with ViT-B-32 obtains an mR of 52.08, which is also slightly lower than EBAKER. It can be mainly attributed to two factors. First, EBAKER is trained with additional data. The larger and more diverse training data may provide EBAKER with broader semantic coverage and improve its robustness. Second, RSITMD contains relatively coarse-grained textual descriptions. FTIR is specifically designed for fine-grained semantic extraction and alignment. Its advantages are less active on RSITMD. Compared to methods based on synthetic data, our proposed method outperforms the previous excellent method GeoRSCLIP [17] by 1.51% in the mR metric. In addition to the quantitative analysis, in order to more intuitively reflect the improvement of our method, we show in Fig. 7 some cases where our proposed FTIR method outperforms RemoteCLIP on the RSITMD test set. We compare FTIR with the superior method RemoteCLIP [18]. FTIR shows better performance in capturing fine-grained and complex semantics, such as quantity and multientity relationships. As shown in the first case of Fig. 7, the query text specifies “five baseball fields.” RemoteCLIP fails to capture the quantity correctly. The top-ranked image contains only three baseball fields, while the ground-truth image appears at rank 3. In contrast, FTIR precisely retrieves the image containing five baseball fields. Moreover, FTIR exhibits superior capability in handling queries involving multiple entities and complex relationships. As shown in the second case of Fig. 7, the query includes parking lots, eight circular reservoirs, roads, and bare lawns. FTIR successfully captures the semantics of multiple entities and their intricate relationships, achieving accurate cross-modal alignment. In contrast, the RemoteCLIP method fails to align some key information. In the RemoteCLIP results, the rank-1 image shows an incorrect number of reservoirs, while the rank-2 image does not contain any parking lots. Notably, the rank-4 image represents an obvious failure case, which shows a square region that contains no reservoirs at all.

3) Performance on the Proposed RSITMD-FG: We have reproduced five representative and competitive methods as

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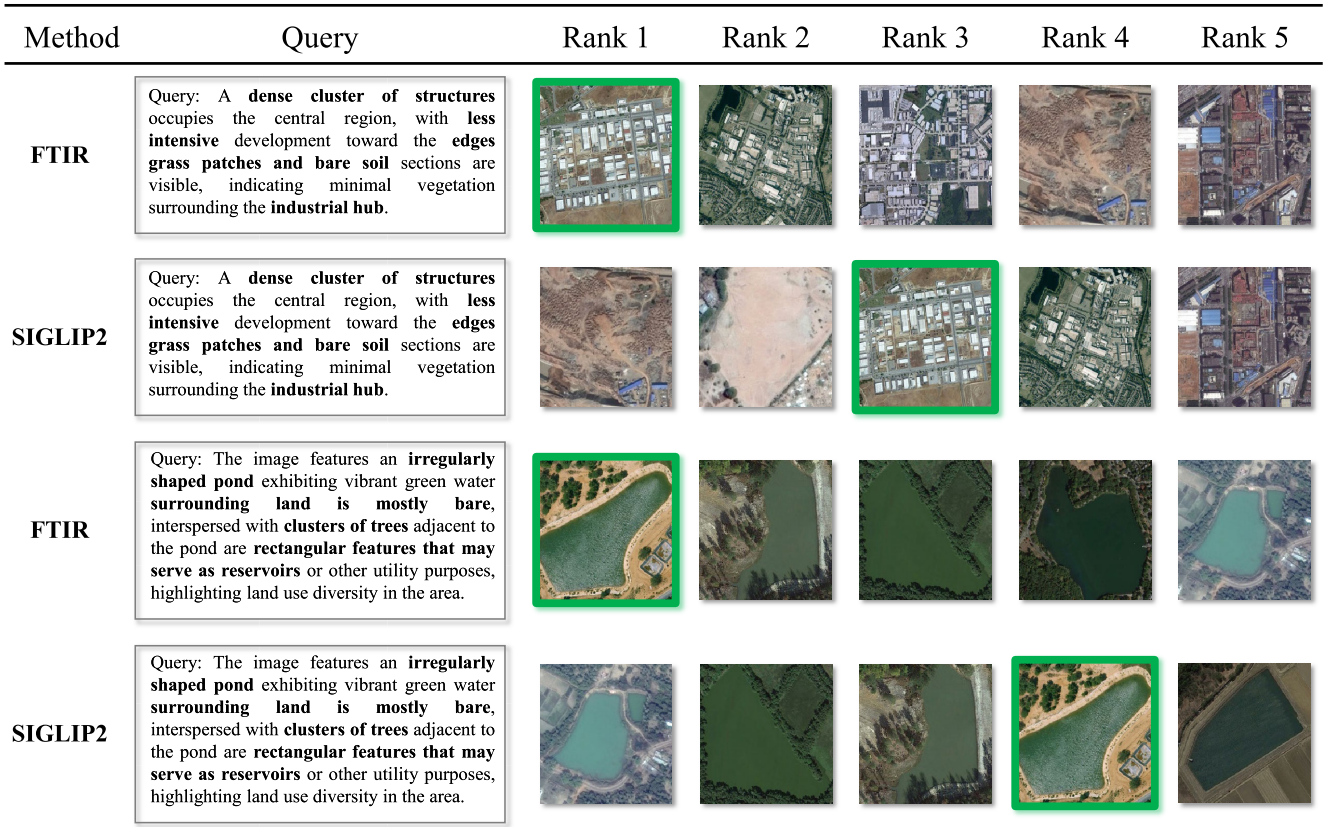


Fig. 8. Comparison of text-to-image retrieval results on the proposed RSITMD-FG dataset. Fine-grained semantic information in the query text is highlighted in bold. Images enclosed in green boxes represent the ground truth. The retrieved images are ranked in descending order of similarity.

baselines. All methods adopt the same optimizer, training data, and number of training epochs. Since EBAKER does not provide publicly available weights, we train it only on the training set of RSITMD-FG. Other models are fine-tuned from the publicly available pretrained weights. Compared with the previous excellent method, CLIP, FTIR achieves an 18.95% improvement in the mR metric. Relative to the baseline method SIGLIP2, FTIR improves the R@1 performance in the text-to-image retrieval task by 2.08%. Furthermore, FTIR outperforms recent advanced models tailored for RS, such as EBAKER and the large-scale RemoteCLIP (ViT-L-14). Specifically, it yields mR improvements of 25.03% and 13.56% over these two methods, respectively. FTIR with ViT-B-32 achieves an mR of 85.18, also yielding an mR improvement of 22.91% over EBAKER. EBAKER mainly emphasizes weak-pair elimination and keyword-level reasoning, which are effective on coarse-grained benchmarks such as RSITMD [15]. By contrast, FTIR is designed for fine-grained semantic extraction and hierarchical alignment. Therefore, under the richer fine-grained supervision of RSITMD-FG and the unified training protocol adopted in our experiments, FTIR exhibits a clear advantage. Benefiting from the proposed RAM module and multigranularity alignment strategy, our method more effectively captures fine-grained semantics in RS images, achieving the best performance on both R@1 and R@5 metrics. In addition, several representative cases in which FTIR outperforms the baseline methods are illustrated in Fig. 8. When performing fine-grained semantic retrieval on the RSITMD-FG dataset, FTIR exhibits a stronger ability to associate the fine-grained semantics in the query text with the corresponding key regions in the image. Moreover, benefiting from the multigranularity alignment strategy, FTIR effectively reduces scene-level mismatches. The first case in Fig. 8 presents the retrieval of an industrial scene. The query description highlights several key elements, including a dense cluster of buildings, grass patches, and areas of bare soil. Functionally, these components collectively characterize an industrial area. Spatially, the buildings are concentrated in the central region, with grass and bare land areas distributed in the surrounding areas. The SIGLIP2 method exhibits obvious scene-level mismatches. In the results obtained from SIGLIP2, both the rank-1 and rank-2 images correspond to the bare land scene, which is unrelated to the queried industrial scene. These mismatches can be attributed to the limited capability of SIGLIP2 in extracting fine-grained semantic information, resulting in the dominance of bare land features during retrieval. FTIR can reduce such mismatches through scene-level alignment in the multigranularity alignment strategy. The proposed FTIR method achieves precise retrieval performance. In the FTIR results, both the rank-2 and rank-3 images correspond closely to the industrial scene characterized by densely clustered buildings. It demonstrates the necessity of the multigranularity alignment strategy. The second case in Fig. 8 concerns the retrieval of a pond scene. It is worth noting that rectangular reservoirs in the query text occupy a small proportion of the image, which is challenging for the

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TABLE VI
COMPARISON OF MODEL EFFICIENCY

Method	Visual Backbone	Parameters	ET (s)
AMFMN [22]	ResNet18	37M	5.11
RemoteCLIP [18]	ViT-B-32	151M	10.07
RemoteCLIP [18]	ViT-L-14	427M	26.73
EBAKER [15]	ViT-B-32	184M	12.58
BLIP [62]	ViT-B-16	223M	50.97
SIGLIP [49]	ViT-B-16	203M	20.45
BLIP2 [63]	ViT-L-14	474M	130.61
SIGLIP2 [50]	ViT-B-16	375M	22.29
FTIR(OURS)	ViT-B-16	465M	28.03

baseline SIGLIP2 method that directly extracts global image features. By employing the RAM module, FTIR establishes fine-grained associations between key regions in the image and the corresponding textual words. FTIR not only accurately extracts the pond information but also captures the fine-grained semantic detail of rectangular reservoirs.

RSITMD-FG shares the same image data and training split with RSITMD. On the text-to-image retrieval task, our proposed FTIR achieves a 34.86% improvement in the R@1 metric on RSITMD-FG compared to RSITMD. This result strongly supports the necessity of fine-grained semantic retrieval and the construction of fine-grained benchmark datasets.

4) Efficiency Comparison: Table VI compares the parameter size and evaluation time of different models. The experiments are conducted on a hardware platform equipped with an Intel Xeon Gold 5318Y CPU running at 2.10 GHz and two NVIDIA A20 GPUs, each with 24 GB of memory. All experiments are implemented on the same platform using PyTorch [57]. Consistent with previous studies [16], [20] [22], we adopt the evaluation time as a metric to measure the time complexity of different methods. ET denotes the evaluation time over the complete test set of RSITMD-FG. The reported evaluation time includes only the online inference cost for all methods. For FTIR, the region features are preextracted by SAM offline and stored in advance, and thus, the offline preprocessing time is not included in this comparison. Although FTIR has a relatively large parameter scale of 465M, it maintains competitive inference efficiency. In particular, compared with BLIP2, which has a similar parameter scale of 474M, FTIR reduces the evaluation time from 130.61 to 28.03 s. This suggests that the proposed RAM in FTIR introduces less computational overhead than the cross-modal interaction design in BLIP2. Although lighter models such as AMFMN and RemoteCLIP require less evaluation time, they show clearly weaker performance on fine-grained retrieval. On RSITMD-FG, FTIR achieves the best overall performance, with an mR of 87.30% and a text-to-image R@1 of 63.93%, outperforming strong baselines such as SIGLIP2. These results suggest that the additional parameters and computational overhead introduced by FTIR are acceptable, given its stronger ability to model fine-grained semantic correspondences. These results also indicate that FTIR achieves a better balance between inference efficiency and retrieval performance.

D. Visualization

For qualitative analysis, we have conducted two main types of visualization analyses to demonstrate the effectiveness of the RAM module and the multigranularity alignment strategy.

1) Visualization of the Proposed RAM: To intuitively demonstrate the effectiveness of the RAM module, we visualize the fine-grained correspondences between key regions in the image and individual words in the text. Specifically, for each sample pair, we average the attention weights across different heads when computing the region-aware features. For each word, the region index with the highest attention weight is taken as the primary region corresponding to that word. According to the weights produced by the semantic importance gating mechanism, we select the top five words. Since no annotations specify associations between words and specific regions, FTIR learns such correspondences in an unsupervised manner. Therefore, these visualizations aim to demonstrate the RAM module's capability to identify meaningful fine-grained associations, rather than to provide precise alignments. We randomly select four samples from the RSITMD-FG dataset and visualize only the correctly learned fine-grained alignments in Fig. 9.

We find that RAM is able to learn fine-grained correspondences between certain important regions and text segments in an unsupervised manner, as we do not explicitly introduce any association information between important regions and text. As shown in Fig. 9(a), the fine-grained semantic information of "tennis court" and "rectangular swimming pool" in the RAM module is accurately aligned with the corresponding key regions in the image. Similarly, as shown in Fig. 9(b), fine-grained semantic concepts such as parts of curved residential structures and the swimming pool in the image receive relatively high attention weights. As shown in Fig. 9(c) and (d), for single distinct subjects such as "a winding bridge" and "circular domed structure," RAM is able to enhance the correspondence between multimodal semantic representations.

2) Visualization of the Proposed MSA Strategy: Regarding the multigranularity alignment strategy, we have verified FTIR's ability to extract semantics at different granularities. Specifically, inspired by SeLo [64], we evaluate FTIR's localization capability for different levels of semantic granularity within a 3000×3000 RS image, as shown in Fig. 10(a). Following SeLo, we slice the RS image into fixed-resolution patches and compute the similarity between each patch and the query text. The overall similarity between the entire image and the query text is then obtained and visualized as a heatmap, as shown in Fig. 10(b).

We find that FTIR is able to effectively perceive semantic information at different granularities. We utilize semantic queries at the scene-level, keyword-level, and fine-grained level, respectively. As shown in Fig. 10(b), when querying the "parking lot" scene, FTIR is able to accurately retrieve all parking lots within the entire image. When querying with the keywords "empty parking lot and empty rows," the parking lot on the left side of the image demonstrates higher semantic similarity. The similarity scores for the fully occupied parking lots

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A bright red [34] tennis court is prominently situated in this image, bordered by neatly trimmed grass. To one side, a [18] rectangular swimming pool stands out, complemented by several parked cars along the adjacent road.

(a)



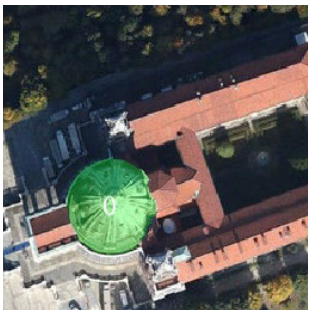
[12] Curved residential structures line the edges of a dark waterway, with several boats moored nearby. The complex features a central [2] swimming [2] pool, surrounded by lush greenery, creating a serene environment.

(b)



Multiple cars are observed traveling on [3] a winding bridge above a river. The surrounding area appears predominantly urban, with a small green patch visible next to the bridge. The water below is a dark, murky color.

(c)



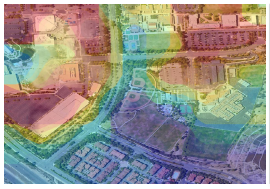
A large, [0] circular domed structure with intricate details is central to the image. Surrounding it are red-tiled roofs forming a rectangular layout. Green trees offer contrast in the background, while a small circular garden is visible within the courtyard

(d)

Fig. 9. Images on the left display the original images from the RSITMD-FG dataset, the corresponding mask regions, and the indices of these masks. On the right, the fine-grained query texts are presented, with important text segments identified by the gating mechanism highlighted using dashed boxes. The labels of the dashed boxes correspond to the indices of the masks with the highest attention scores. (a) Tennis court and a rectangular swimming pool. (b) Curved residential structures and a swimming pool along a waterway. (c) Winding bridge. (d) Circular domed structure.

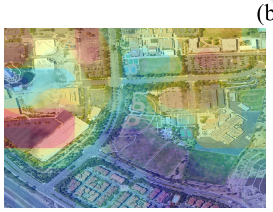


(a)



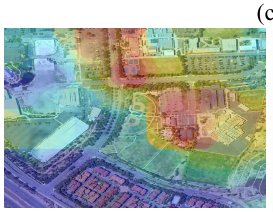
Query: Parking Lot.

(b)



Query: Big **empty** parking lot, many empty rows

(c)



Query: There is a parking lot sparsely populated with **a few cars** and surrounded by **two tennis courts** and **some basketball courts**, the lot provides access to the surrounding sports fields.

(d)

Fig. 10. Semantic perception capability of FTIR across different semantic granularities. (a) 3000×3000-pixel RS image before slicing. (b) Similarity heatmap between image patches and scene-level queries. (c) Similarity heatmap corresponding to keyword-level queries. (d) Similarity heatmap for fine-grained queries. A deeper red color indicates a higher similarity, whereas a deeper blue color denotes a lower similarity.

on the right decrease accordingly, as shown in Fig. 10(c). As shown in Fig. 10(d), when the semantic granularity becomes more detailed, FTIR is able to accurately localize composite semantics such as the combination of tennis courts, basketball courts, and nearby parking lots.

E. Limitations

FTIR has made a preliminary attempt to explore the fine-grained associations between visual and textual modalities in the RSTIR. However, RAM performs fine-grained matching between region-level features and word-level features only

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after extracting global representations from both modalities. Consequently, modality alignment in RAM is conducted at a relatively shallow level rather than through deep fusion during the feature extraction stage. This design limits the joint optimization of visual and textual representations. There is still room for improvement in capturing precise correspondences between regions and words. Future work could further enhance fine-grained cross-modal alignment. It could be achieved by introducing deep fusion mechanisms during feature extraction or incorporating semantic reranking strategies in the matching phase.

Although RSITMD-FG is a high-quality dataset with both images and query texts at a fine-grained level, its scale is only 23K image–text pairs. It is relatively small compared with other large-scale datasets. Inspired by the scaling law, increasing the amount of data could further enhance FTIR's ability to perceive fine-grained semantic information. Therefore, we plan to expand the scale of RSITMD-FG to provide a more comprehensive benchmark for the RSTIR field.

VI. Conclusion and Future Work

In this work, we have proposed the FTIR framework to capture fine-grained correspondences between visual regions and textual semantics. Our method incorporates the RAM module to enhance the perception of fine-grained semantics. Then, we employ a multigranularity alignment strategy to reduce the difficulty and noise sensitivity of fine-grained semantic alignment. Furthermore, we construct a high-quality fine-grained dataset, RSITMD-FG, which provides detailed region-level annotations and fine-grained textual descriptions.

Extensive experiments demonstrate that our proposed FTIR achieves competitive performance across multiple benchmarks, surpassing even some domain-specific foundation models. It remains concise without using complex multimodal interactions or data augmentation. As discussed above, FTIR still has several limitations. Cross-modal features in RAM are aligned at a relatively shallow level. There is still room for capturing precise correspondences between regions and words. Although RSITMD-FG comprises 23K high-quality sample pairs, its overall scale remains limited and could benefit from further expansion.

In future research, we plan to extend our work to multiturn interactive retrieval tasks, enabling more dynamic and context-aware retrieval scenarios. In addition, we aim to further explore how to effectively leverage the powerful generative capabilities of large vision-language models to enable deep fusion of fine-grained semantic representations. We also hope to incorporate semantic reranking strategies at the matching stage for RSTIR.

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